Reagent Probe Position Adjustment

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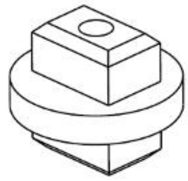
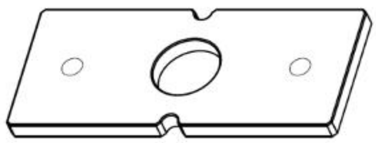
1. Adjustment tool

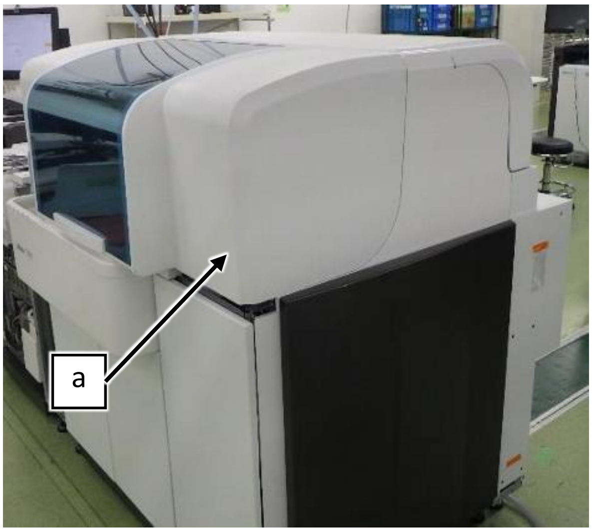
The following adjustment tools are used for this adjustment. Please refer to "S03-01 List of adjustment tools" for a list of jigs.

P/N	Name	Appearance
791-1831	R CELL ADJUSTING TOOL	
792-6453	BOTTLE PLATE C	 X4

2. Preparation

2.1 Removing the covers

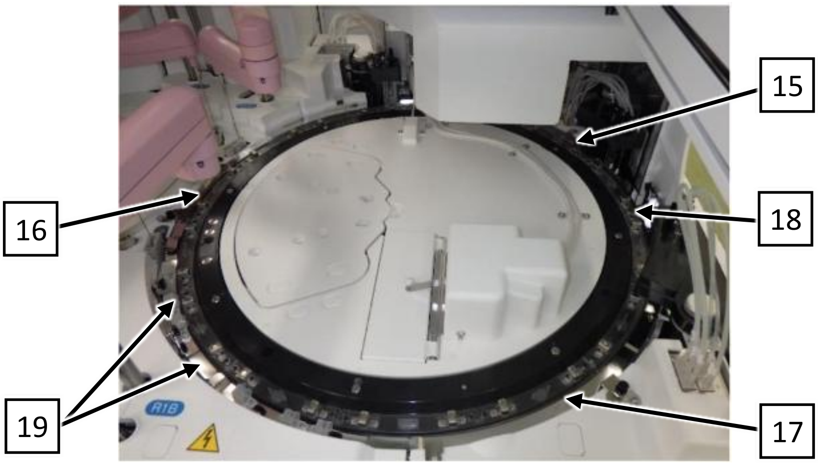
(1) Open TOP COVER F ASSY (a).



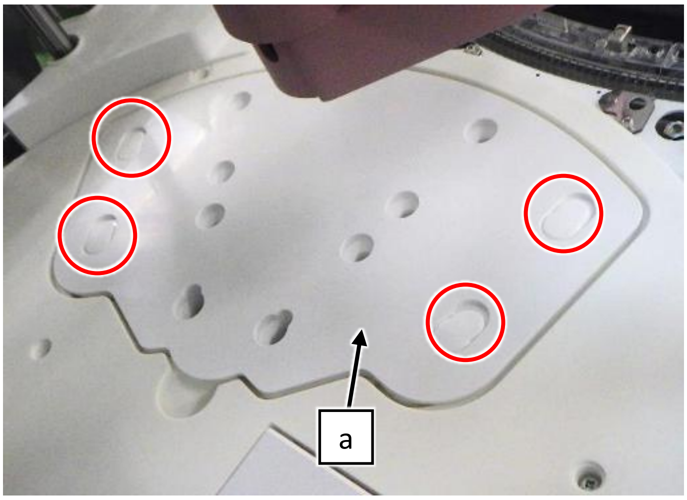
(2) Loosen the screw and remove COVER MSW ASSY (a).

(3) Turn on Maintenance Switch.

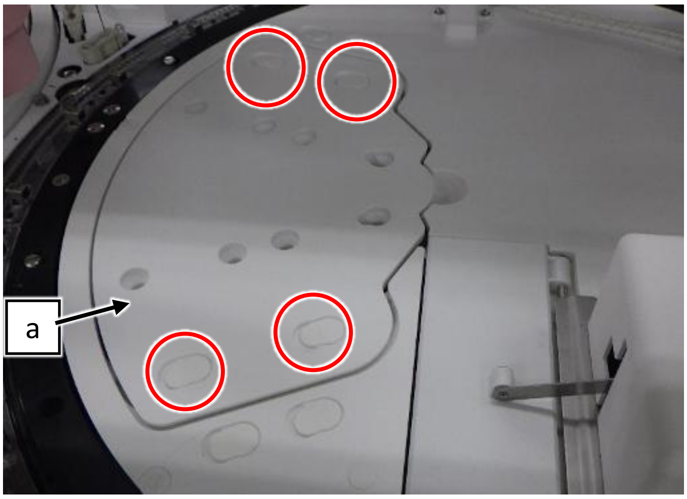
(4) Remove 3 Cell covers (No.15, 16) (refer to “S06-41 Removement and attachment of covers”).



(5) Remove the 4 screw caps and the 4 screws, and then REAGENT PIPETTING COVER (a).



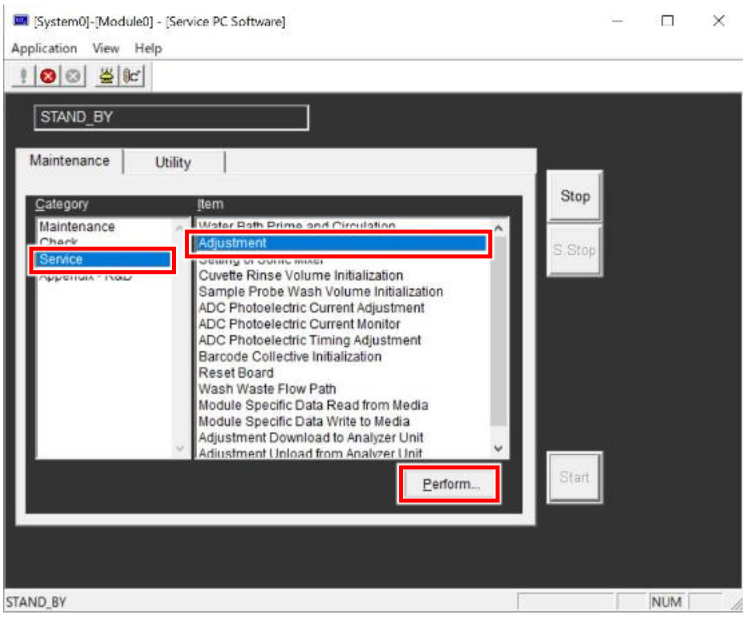
(6) Remove the 4 screw caps and the 4 screws, and then REAGENT PIPETTING COVER (a).



2.2 Start SSW

(1) Refer to “17 pro v2 SSW Adjustment Procedure”.

(2) Select [Service] > [Adjustment] > [Perform...] in SSW.



3. Adjusting Reagent Probe R1-A Horizontal

The adjustment of Reagent Disk A and Reagent Disk B should be completed.

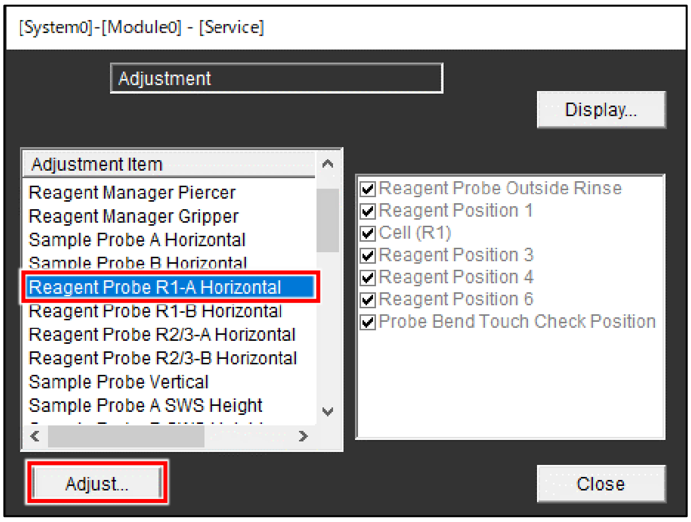
If it hasn't been adjusted yet, perform the adjustments (refer to “S03-08 Reagent Manager Position and Gripper Assy Adjustment”).

The adjustment procedure for R1-B, R2/3-A, R2/3-B are basically same as the procedure for R1-A.

Refer to the following table for the adjustment items for each reagent probe.

Adjustment item	Reagent Probe			
	R1-A	R1-B	R2/3-A	R2/3-B
3.1 Reagent Probe Outside Rinse position	×	×	×	×
3.2 Reagent Position 1	×	×	×	×
3.3 Cell (R1) position	×	×	-	-
5.1 Cell (R2) position	-	-	×	×
3.4 Reagent Position 3	×	×	×	×
5.2 Cell (R3) position	-	-	×	×
3.5 Reagent Position 4	×	×	×	×
3.6 Reagent Position 6	×	×	×	×
3.7 Prove Bend Touch Check Position	×	×	×	×
4.1 Basic Wash Check Position	-	×	-	-

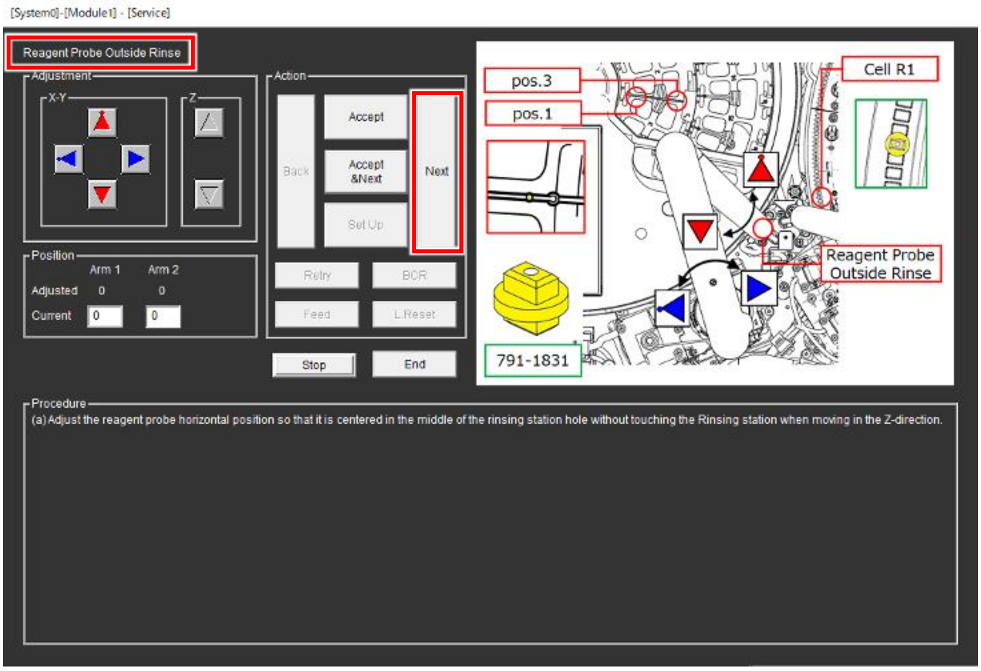
(1) Select [Reagent Probe R1-A Horizontal] from Adjustment item and select [Adjust...].



3.1 Reagent Probe Outside Rinse position

(1) Make sure that the adjustment subject is "Reagent Probe Outside Rinse" at the upper left of SSW.

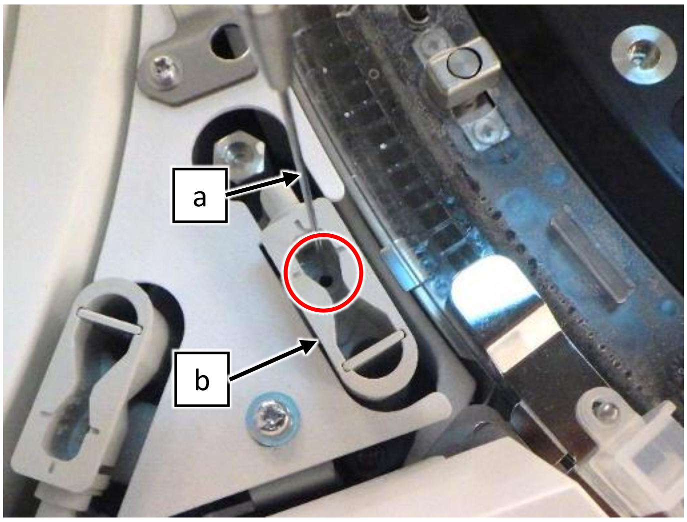
If not, select [Next] repeatedly until "Reagent Probe Outside Rinse" appears.



(2) Lower Reagent probe arm and make sure that the tip of Reagent probe (a) is at the center of the washing hole of the washing station (b).

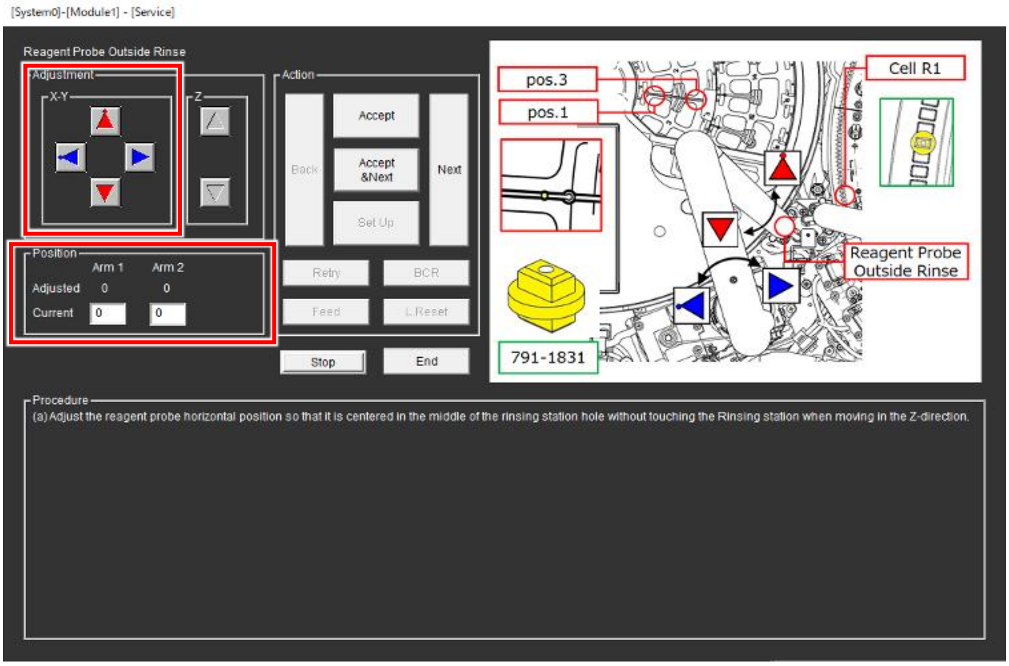
If it is at the center, select [Next] and proceed to the next adjustment.

If not, proceed to the next step.

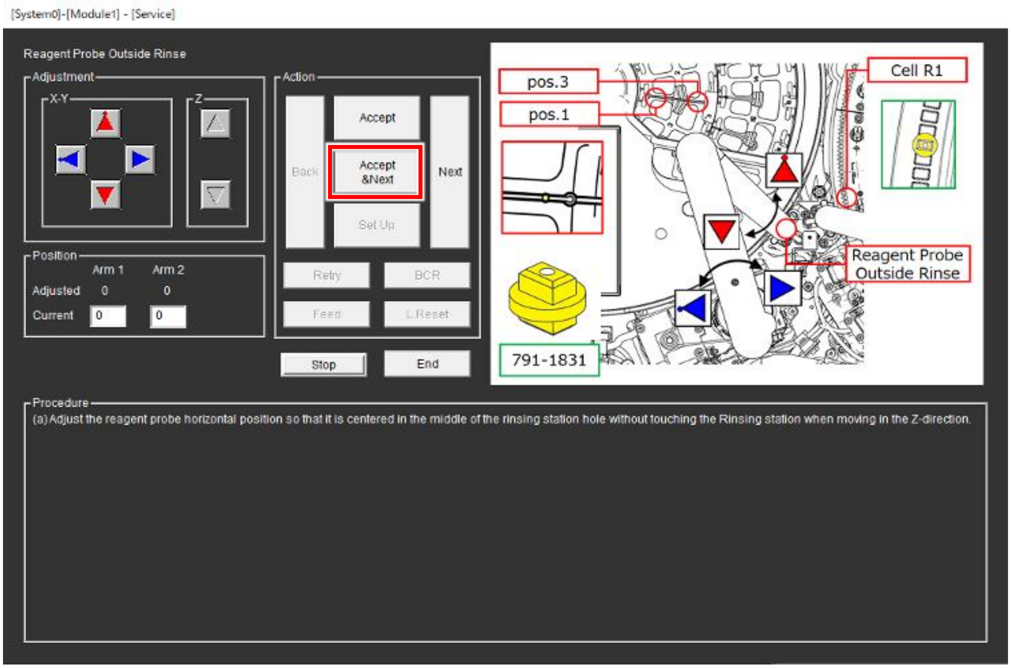


(3) Adjust Reagent probe position so that the tip of Reagent probe is at the center of the washing hole using the 4 arrow buttons.

(Range: Arm 1: ±50 pulse, Arm 2: ±100 pulse)



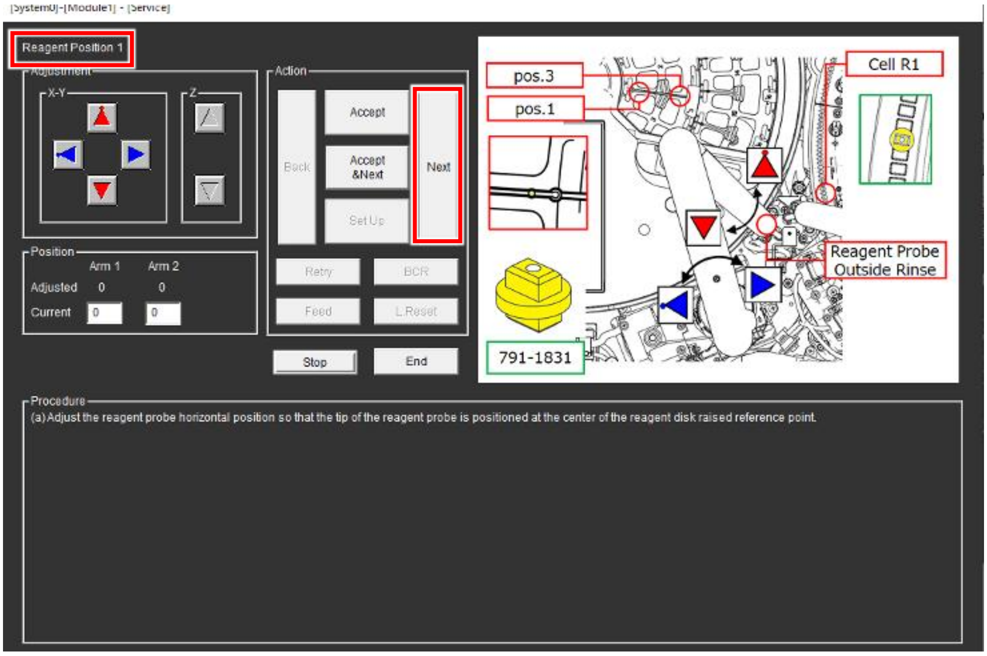
(4) Select [Accept & Next].



3.2 Reagent Position 1

(1) Make sure that the adjustment subject is “Reagent Position 1” at the upper left of SSW.

If not, select [Next] repeatedly until “Reagent Position 1” appears.



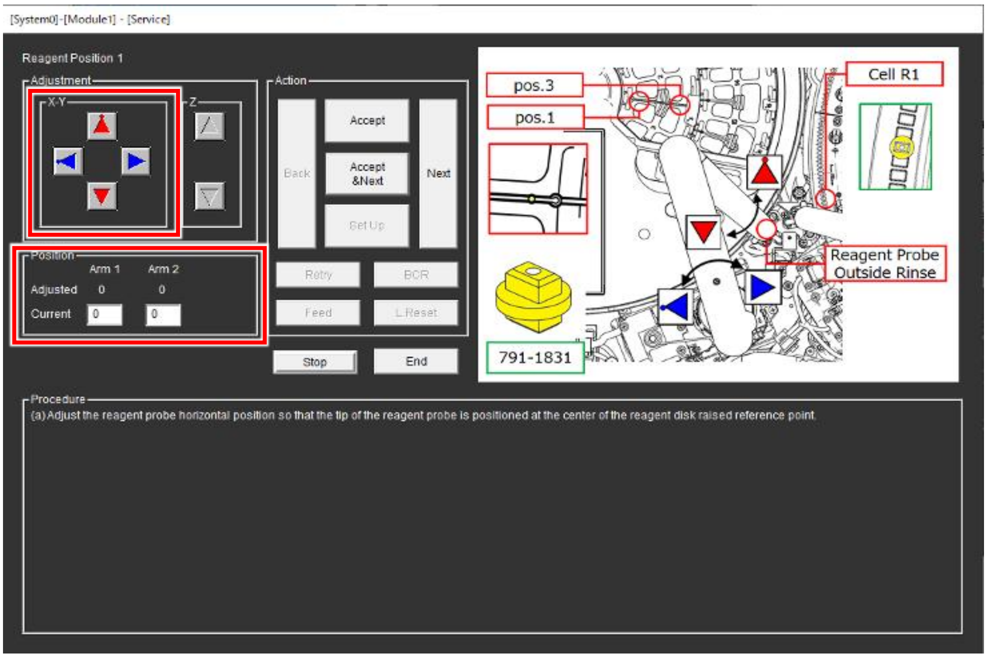
(2) Lower Reagent probe arm and make sure that the tip of Reagent probe touches the protrusion on the rib of Reagent disk.

If it is at the center, select [Next] and proceed to the next adjustment.

if not, proceed to the next step.

(3) Adjust Reagent probe position so that the tip of reagent probe touches the protrusion on the rib using the 4 arrow buttons.

(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)



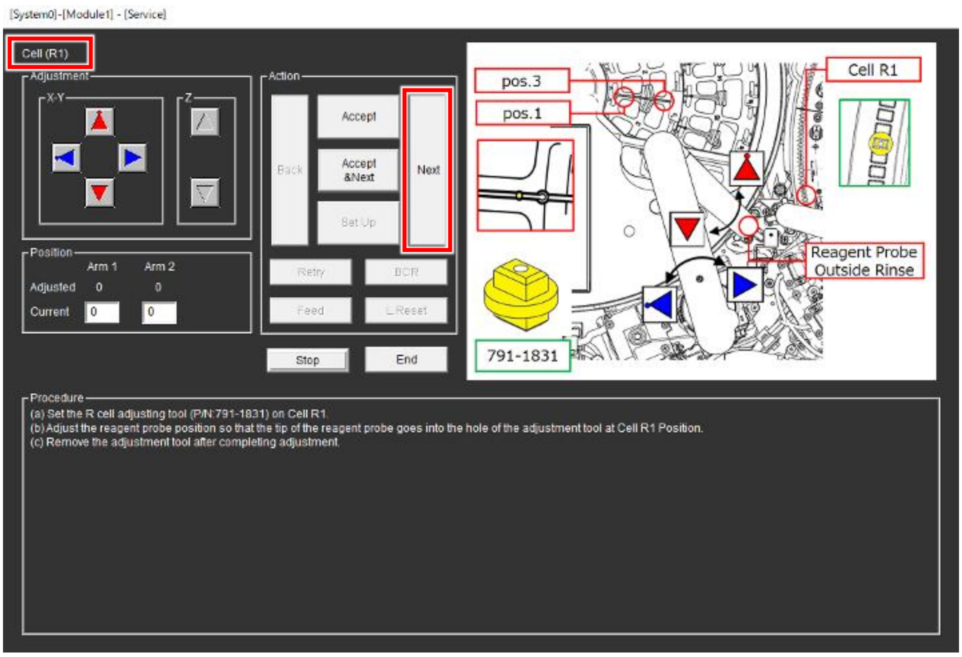
(4) Select [Accept & Next].

3.3 Cell (R1) position

This position is for R1-A and R1-B.

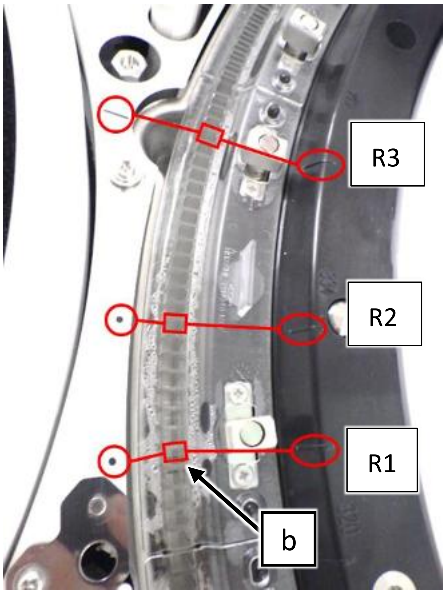
(1) Make sure that the adjustment subject is “Cell (R1)” at the upper left of SSW.

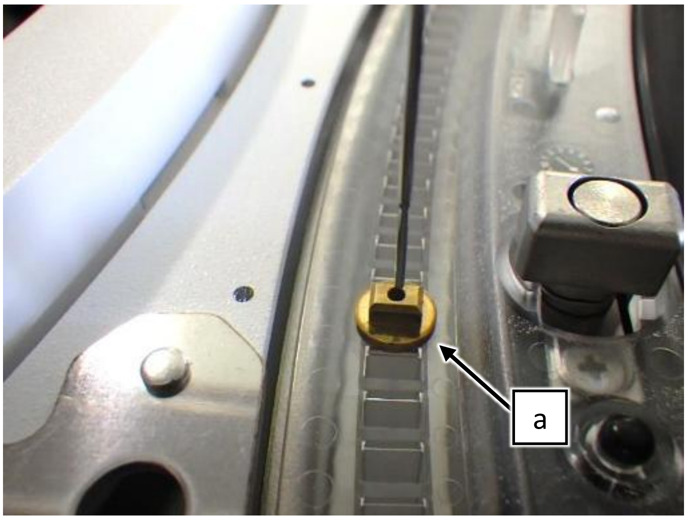
If not, select [Next] repeatedly until “Cell (R1)” appears.



(2) Set the tool (a) on the R1 cell (b) between the dot marker and the line marker.

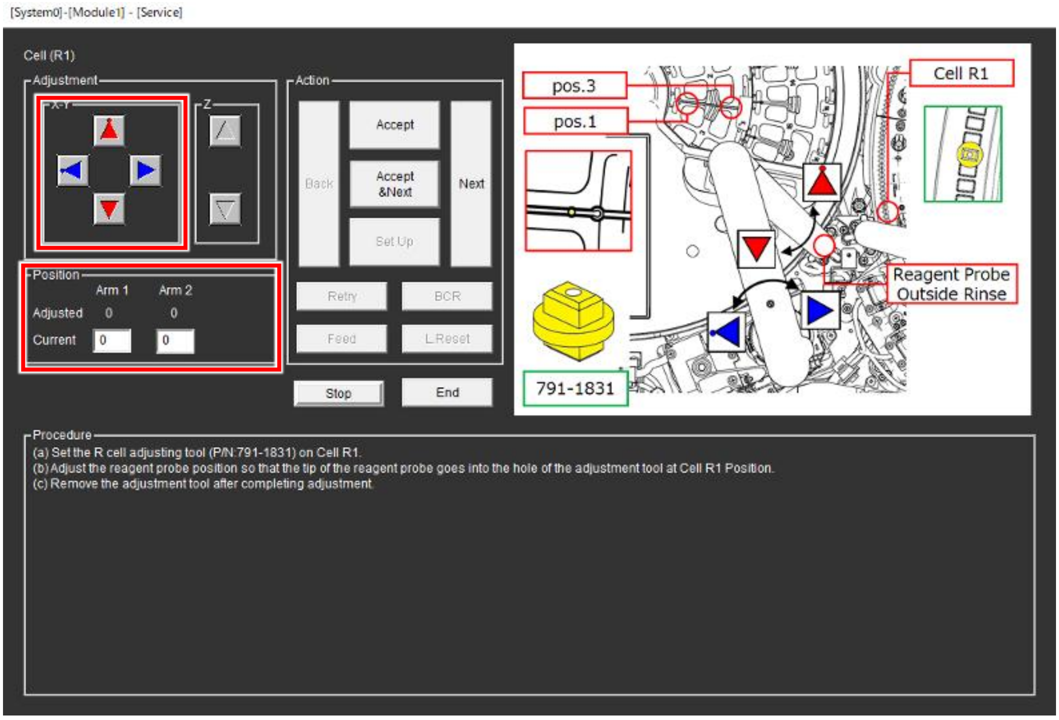
(3) Lower Reagent probe arm and make sure that the tip of Reagent probe is at the center of the tool (a) hole.
If it is at the center, select [Next] and proceed to the next adjustment.
if not, proceed to the next step.





(a) R CELL ADJUSTING TOOL (P/N: 791-1831)

(4) Adjust Reagent probe position so that the tip of Reagent probe is at the center of the tool hole using Use the 4 arrow buttons.
(Range: Arm 1: ±30 pulse, Arm 2: ± 60 pulse)



(5) Select [Accept].

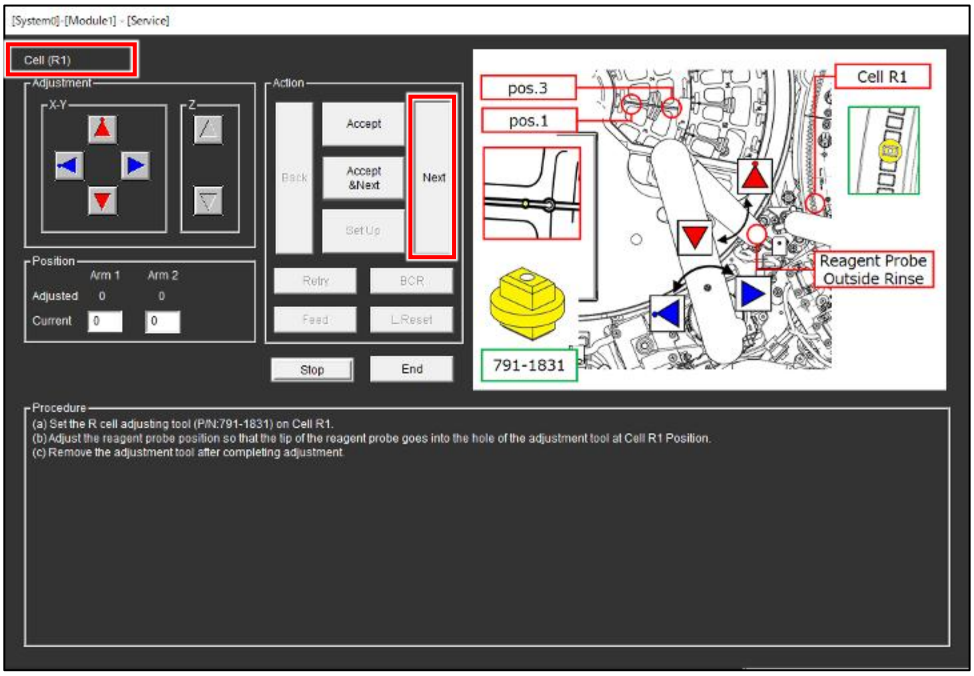
(6) Remove the tool from R1 cell.

(7) Select [Next].

3.4 Reagent Position 3

(1) Make sure that the adjustment subject is “Reagent Position 3” at the upper left of SSW.

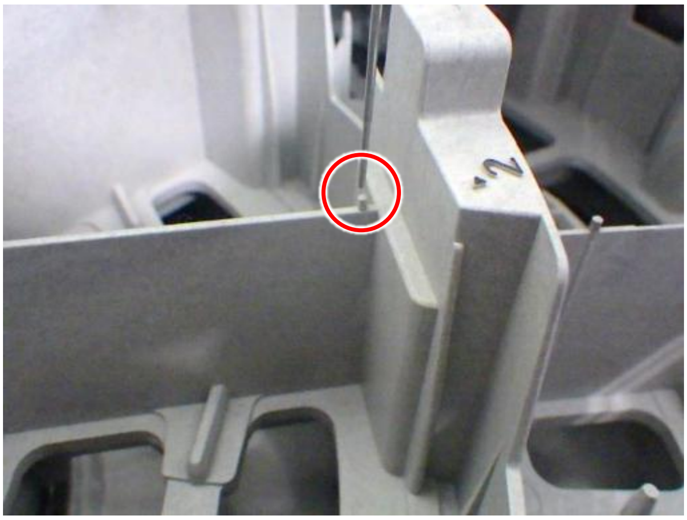
If not, select [Next] repeatedly until “Reagent Position 3” appears.



(2) Lower Reagent probe arm and make sure that the tip of Reagent probe touches the protrusion on the rib of Reagent disk.

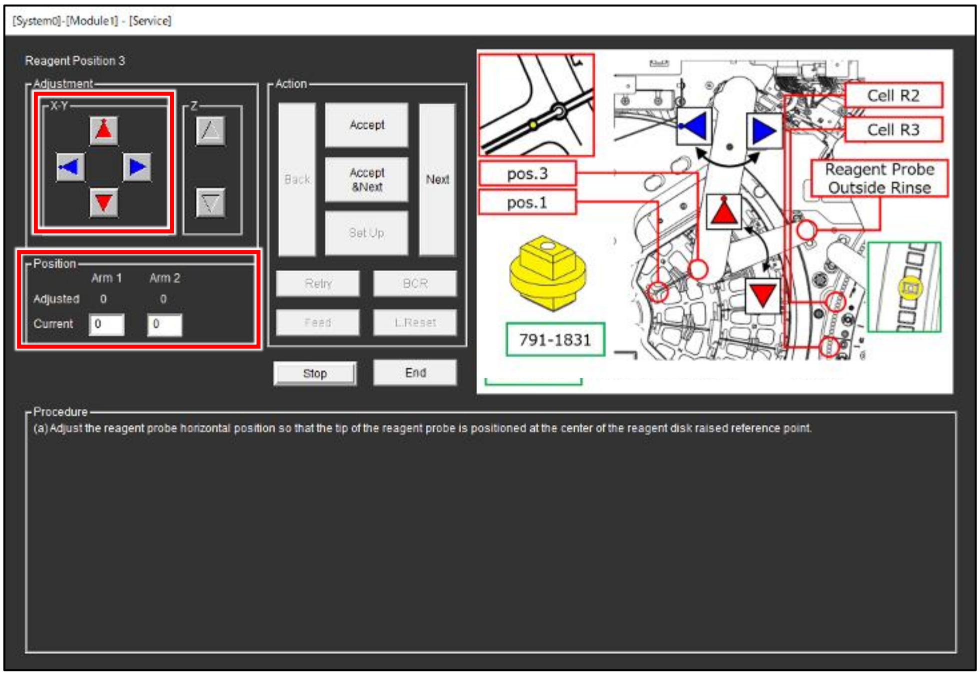
If it is at the center, select [Next] and proceed to the next adjustment.

if not, proceed to the next step.



(3) Adjust Reagent probe position so that the tip of reagent probe touches the protrusion on the rib using the 4 arrow buttons.

(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)

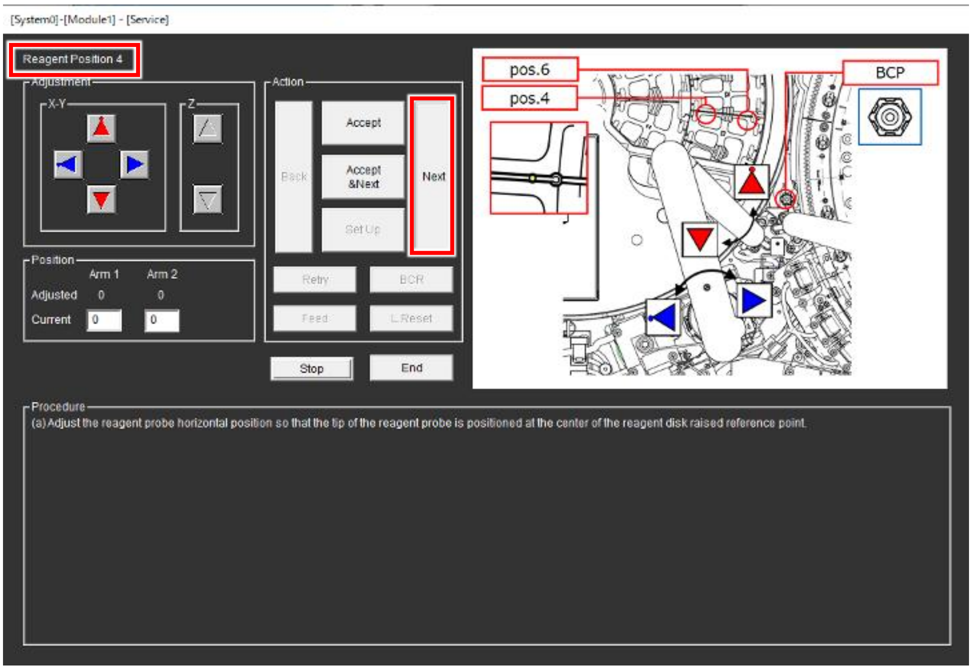


(4) Select [Accept & Next].

3.5 Reagent Position 4

(1) Make sure that the adjustment subject is “Reagent Position 4” at the upper left of SSW.

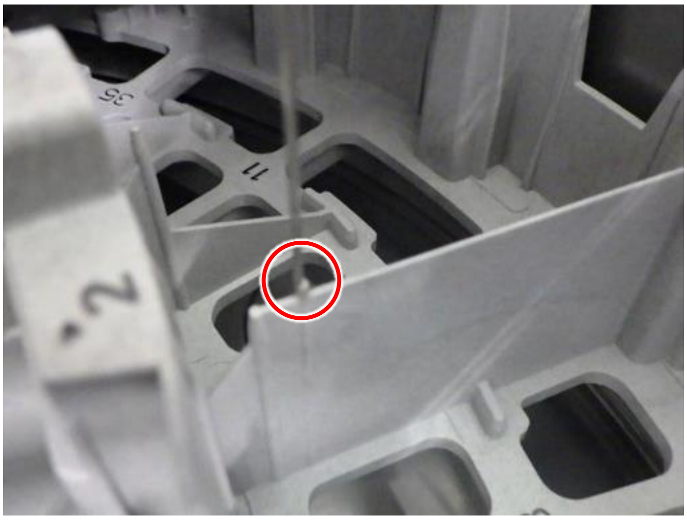
If not, select [Next] repeatedly until “Reagent Position 4” appears.



(2) Lower Reagent probe arm and make sure that the tip of Reagent probe touches the protrusion on the rib of Reagent disk.

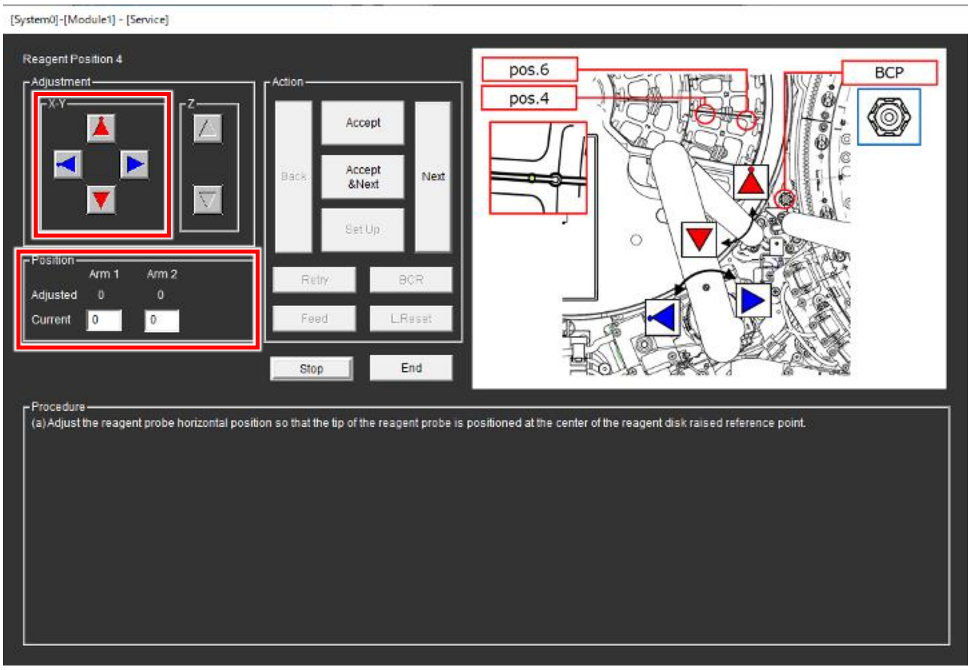
If it is at the center, select [Next] and proceed to the next adjustment.

if not, proceed to the next step.



(3) Adjust Reagent probe position so that the tip of reagent probe touches the protrusion on the rib using the 4 arrow buttons.

(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)

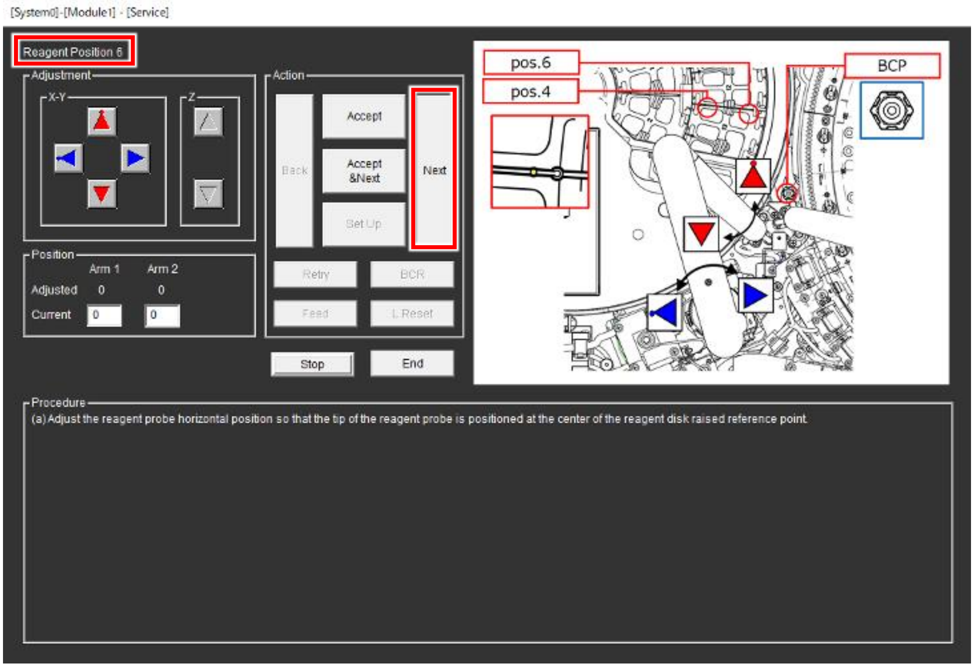


(4) Select [Accept & Next].

3.6 Reagent Position 6

(1) Make sure that the adjustment subject is “Reagent Position 6” at the upper left of SSW.

If not, select [Next] repeatedly until “Reagent Position 6” appears.



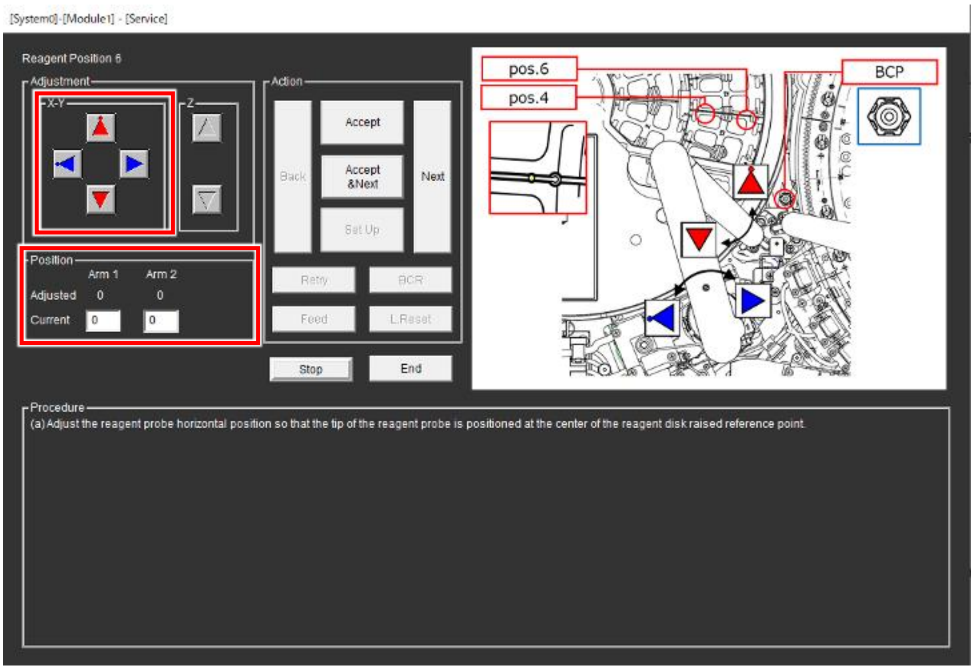
(2) Lower Reagent probe arm and make sure that the tip of Reagent probe touches the protrusion on the rib of Reagent disk.

If it is at the center, select [Next] and proceed to the next adjustment.

if not, proceed to the next step.

(3) Adjust Reagent probe position so that the tip of Reagent probe touches the protrusion on the rib using the 4 arrow buttons.

(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)

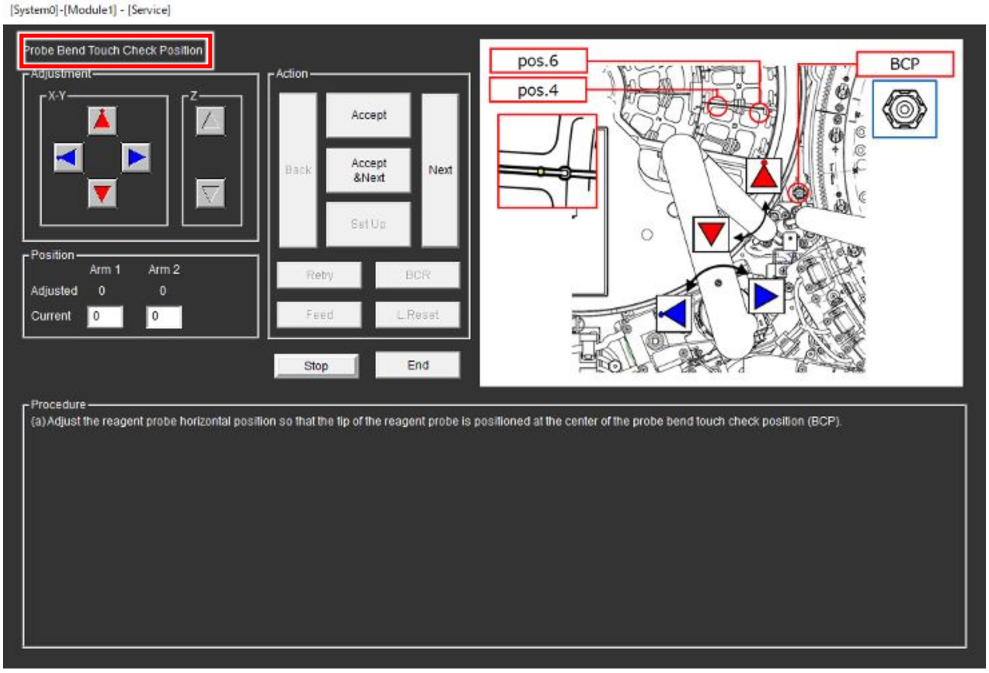


(4) Select [Accept & Next].

3.7 Probe Bend Touch Check Position

(1) Make sure that the adjustment subject is “Prove Bend Touch Check Position” at the upper left of SSW.

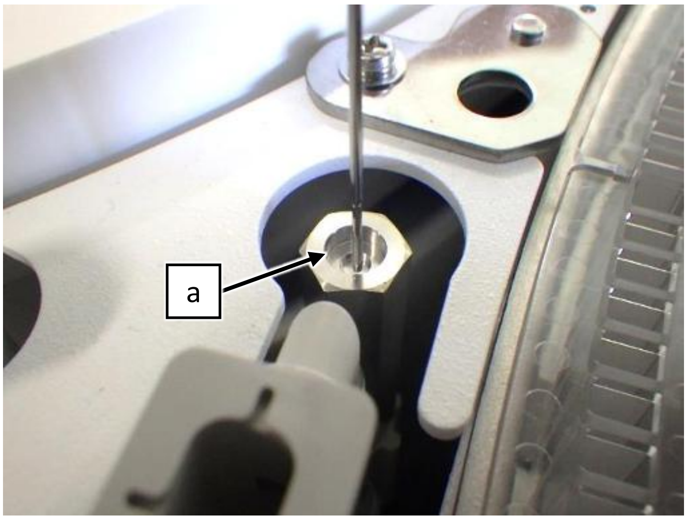
If not, select [Next] repeatedly until “Prove Bend Touch Check Position” appears.



(2) Lower Reagent probe arm and make sure that the tip of Reagent probe is at the center of the block at bend check position (a).

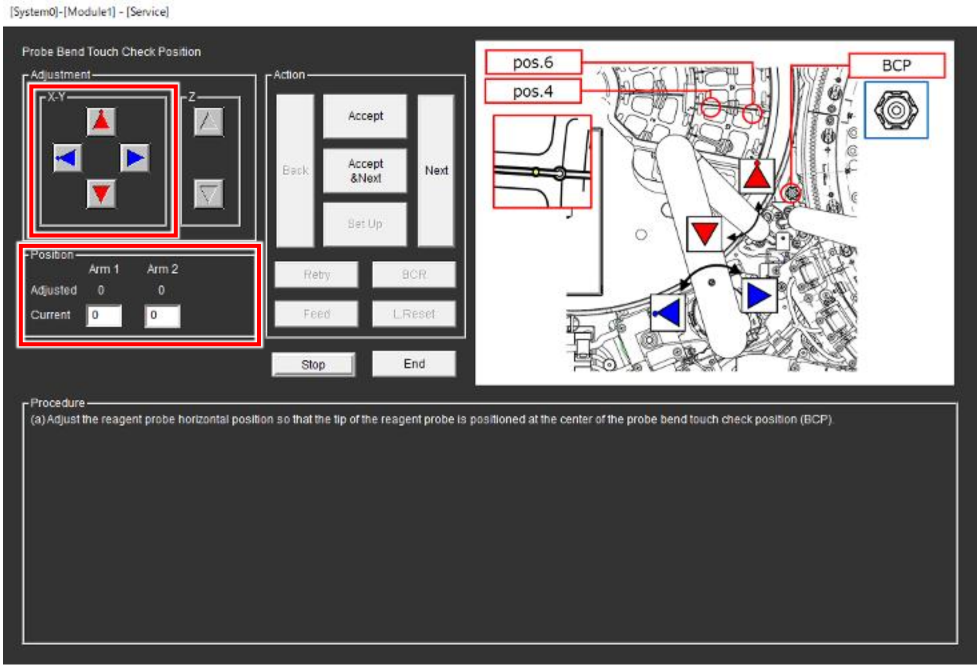
If it is at the center, select [Next] and proceed to the next adjustment.

if not, proceed to the next step.



(3) Adjust Reagent probe position so that the tip of Reagent probe is at the center of the bend check position using the 4 arrow buttons.

(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)



(4) Select [Accept & Next].

4. Adjusting Reagent Probe R1-B Horizontal

The adjustment procedure for R1-B are basically same as the procedure for R1-A.

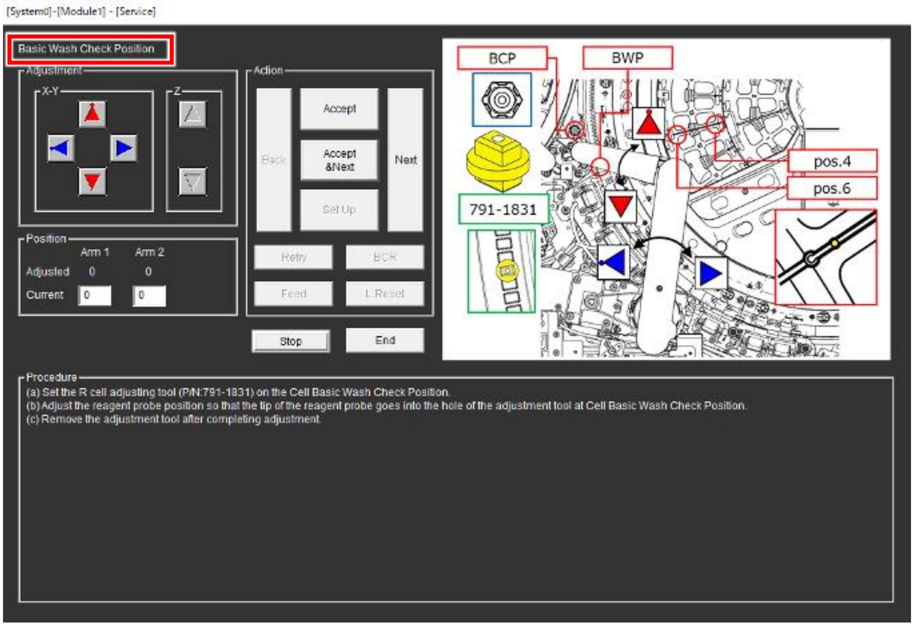
Perform the adjustment refer to the chapter 3.

4.1 Basic Wash Check position

This position is only for R1-B.

(1) Make sure that the adjustment subject is “Basic Wash Check Position” at the upper left of SSW.

If not, select [Next] repeatedly until “Basic Wash Check Position” appears.

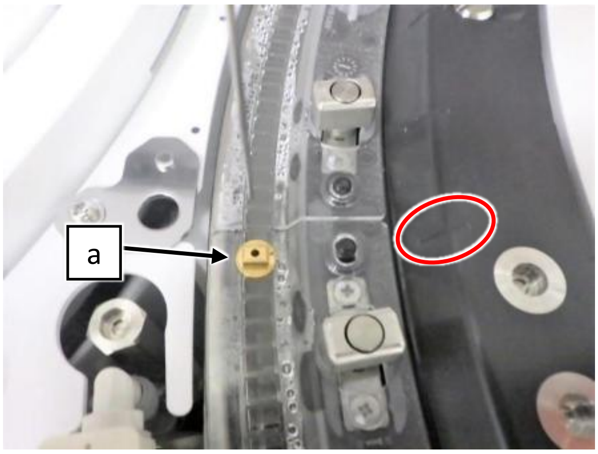


(2) Set the tool (a) on the cell according to the line marker.

(3) Lower Reagent probe arm and make sure that the tip of Reagent probe is at the center of the tool (a) hole.

If it is at the center, select [End] and proceed to the next adjustment.

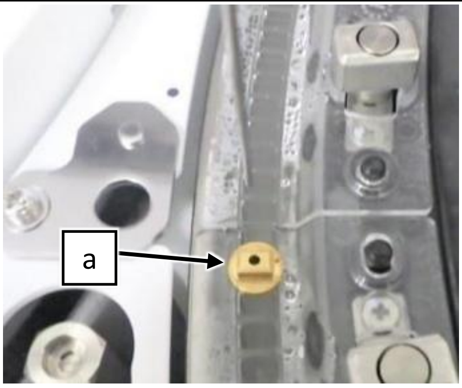
if not, proceed to the next step.



(a): R CELL ADJUSTING TOOL (P/N: 791-1831)

(4) Adjust Reagent probe position so that the tip of Reagent probe is at the center of the tool hole.

(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)



(5) Select [Accept].

(6) Remove the tool from the cell.

(7) Select [Next].

(8) Select [End] after the complete of the adjustment task. (for R1-B)

5. Adjusting Reagent Probe R2/3-A Horizontal

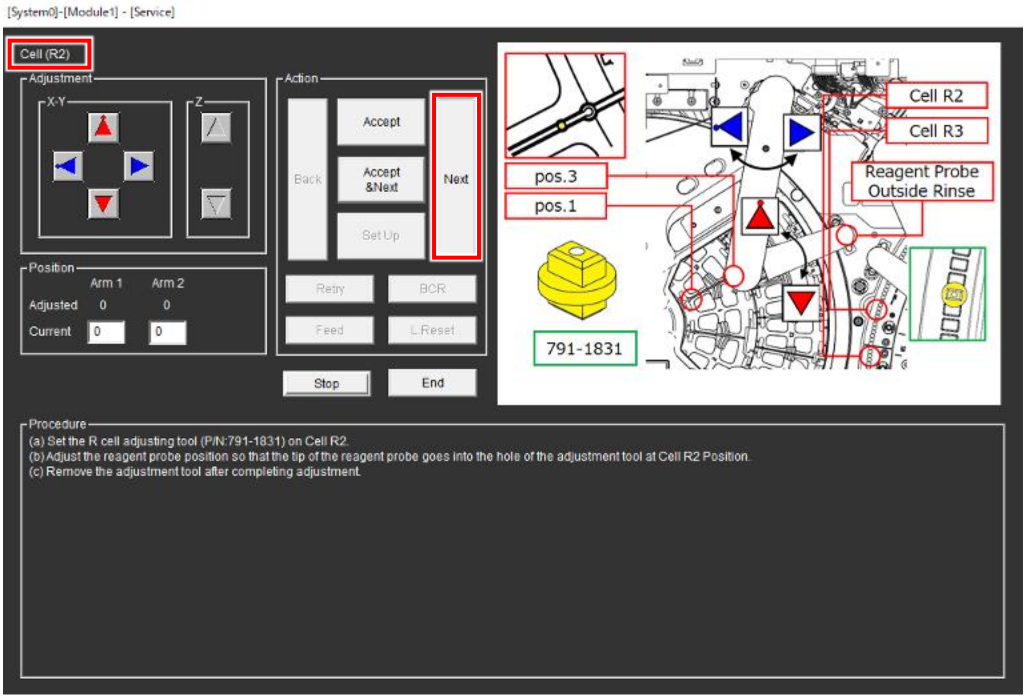
The adjustment procedure for R2/3-A are basically same as the procedure for R1-A.

5.1 Cell (R2) position

This position is for R2/3-A and R2/3-B.

(1) Make sure that the adjustment subject is "Cell (R2)" at the upper left of SSW.

If not, select [Next] repeatedly until "Cell (R2)" appears.

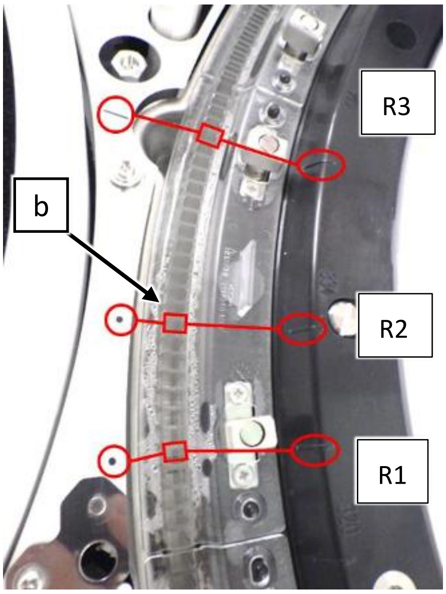


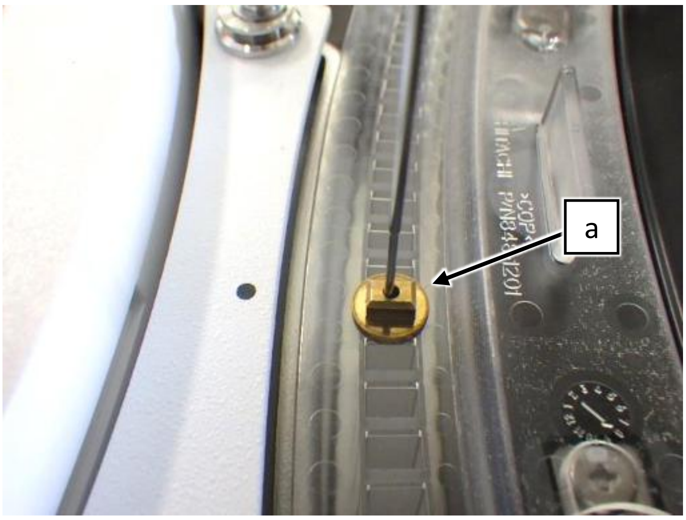
(2) Set the tool (a) on R2 cell (b) between the dot marker and the line marker.

(3) Lower Reagent probe arm and make sure that the tip of Reagent probe is at the center of the tool (a) hole.

If it is at the center, select [Next] and proceed to the next adjustment.

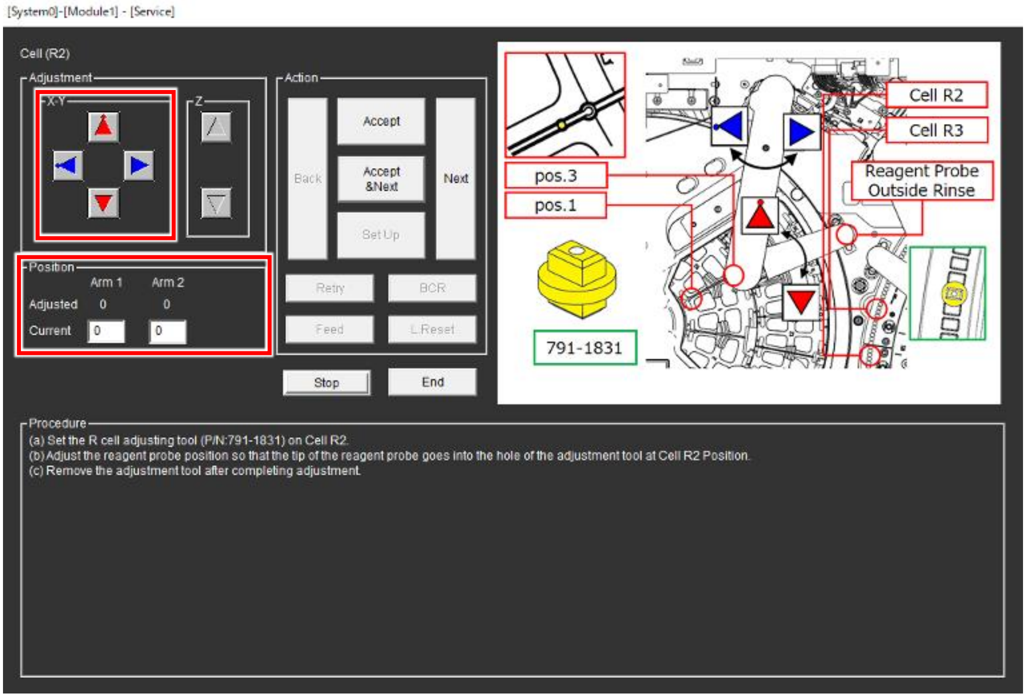
if not, proceed to the next step.





(a): R CELL ADJUSTING TOOL (P/N:791-1831)

(4) Adjust Reagent probe position so that the tip of Reagent probe is at the center of the tool hole using the 4 arrow buttons.
(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)

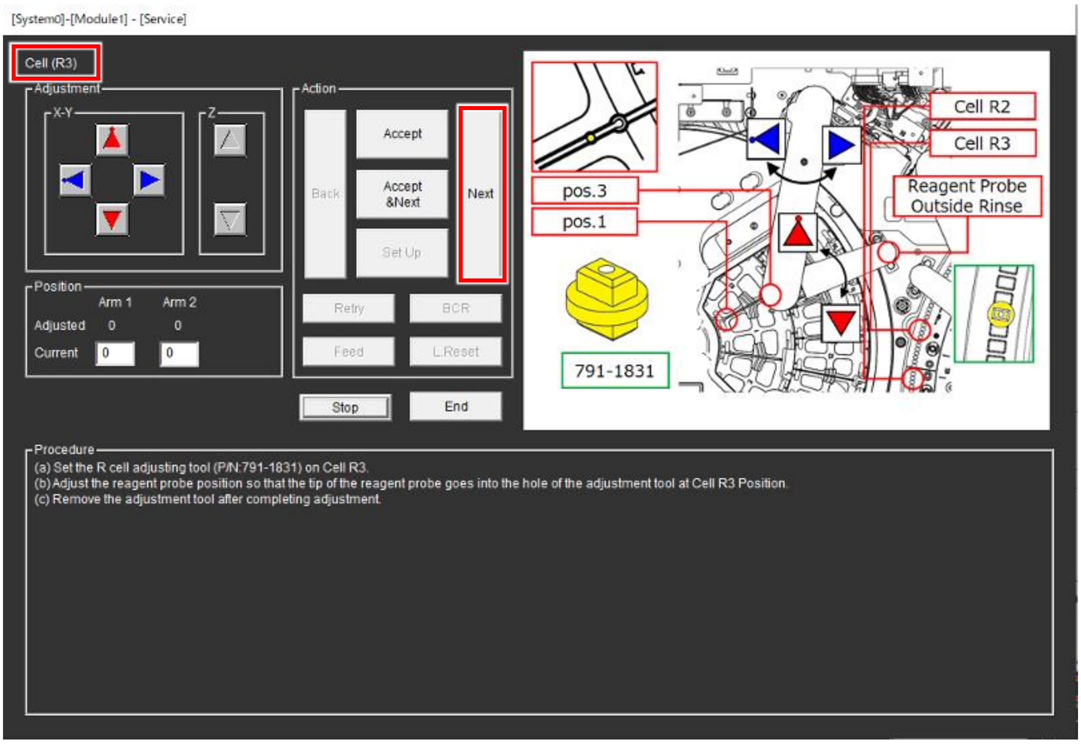


(5) Select [Accept].
(6) Remove the tool from R2 cell.
(7) Select [Next].

5.2 Cell (R3) position

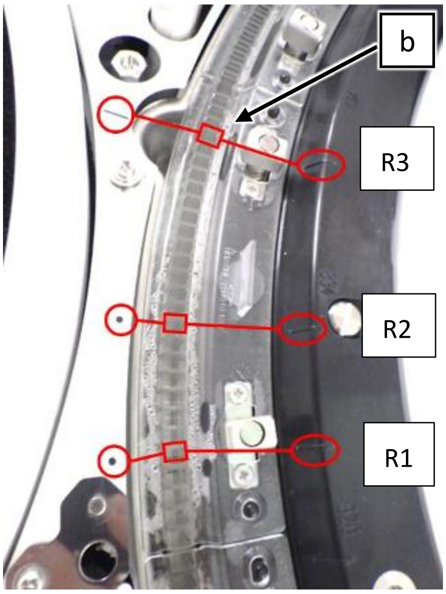
This position is for R2/3-A and R2/3-B.

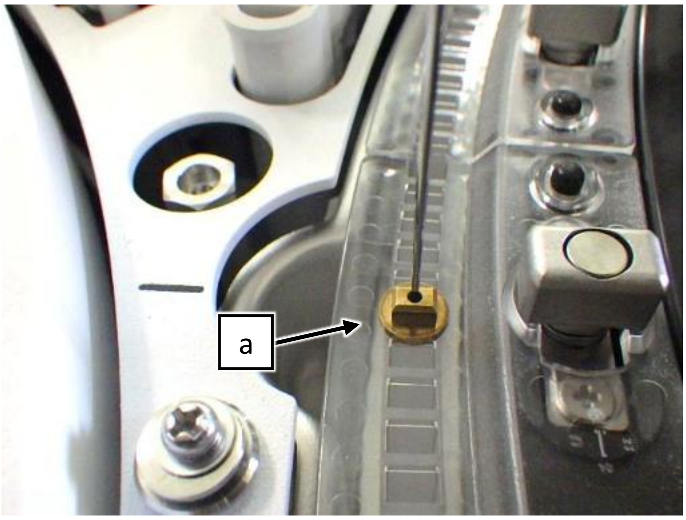
(1) Make sure that the adjustment subject is “Cell (R3)” at the upper left of SSW.
If not, select [Next] repeatedly until “Cell (R3)” appears.



(2) Set the tool (a) on R3 cell (b) between the 2 line markers.

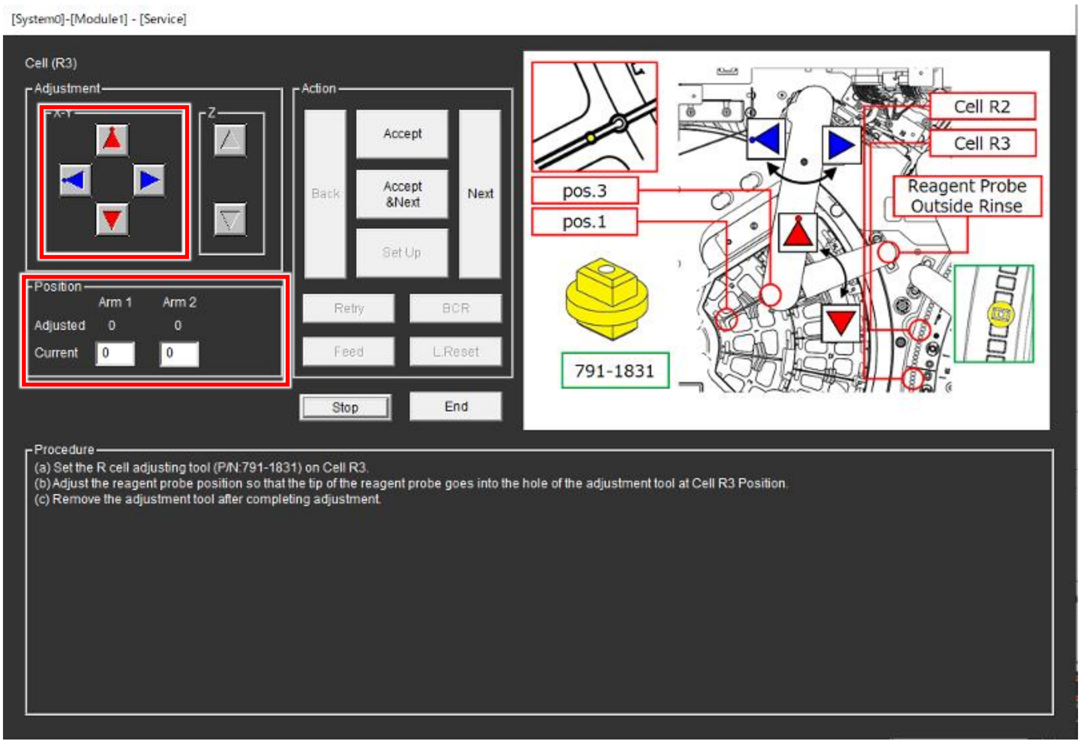
(3) Lower Reagent probe arm and make sure that the tip of Reagent probe is at the center of the tool (a) hole.
If it is at the center, select [Next] and proceed to the next adjustment.
if not, proceed to the next step.





(a) R CELL ADJUSTING TOOL (P/N:791-1831)

(4) Adjust Reagent probe position so that the tip of Reagent probe is at the center of the tool (a) hole using the 4 arrow buttons.
(Range: Arm 1: ±30 pulse, Arm 2: ±60 pulse)



(5) Select [Accept].

(6) Remove the tool from R3 cell.

(7) Select [Next].

6. Adjusting Reagent Probe R2/3-B Horizontal

The adjustment procedure for R2/3-B are basically same as the procedure for R1-A.

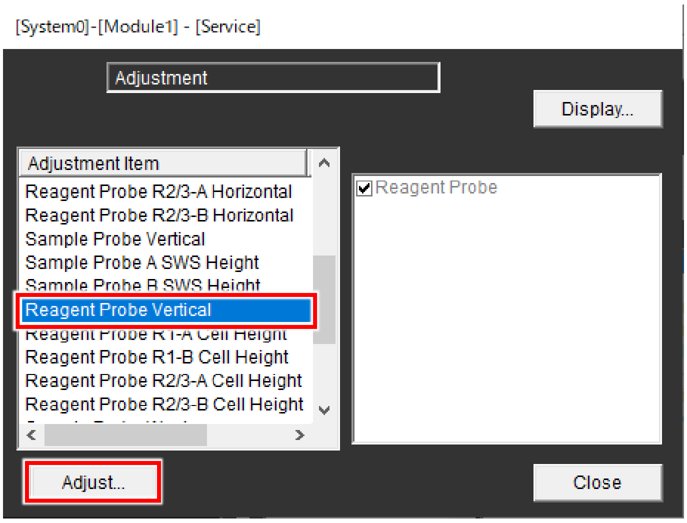
Perform the adjustment refer to the chapter 3.

7. Adjusting Reagent Probe Vertical

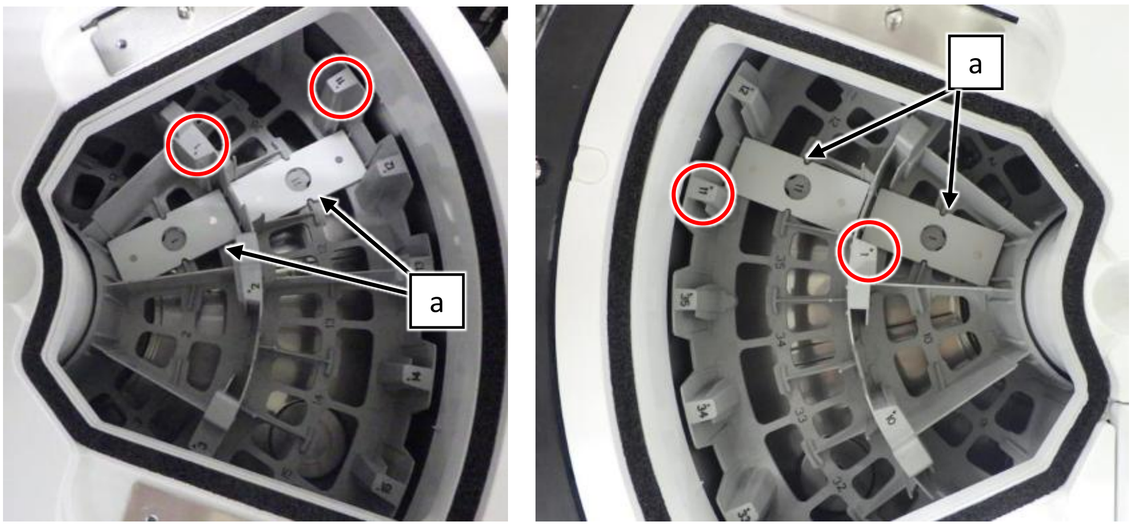
The adjustment of horizontal stop position for the Reagent Probe should be completed.

If it hasn't been adjusted yet, perform the adjustment in accordance with the adjustment procedure.

(1) Select [Reagent Probe Vertical] from Adjustment item and select [Adjust...].

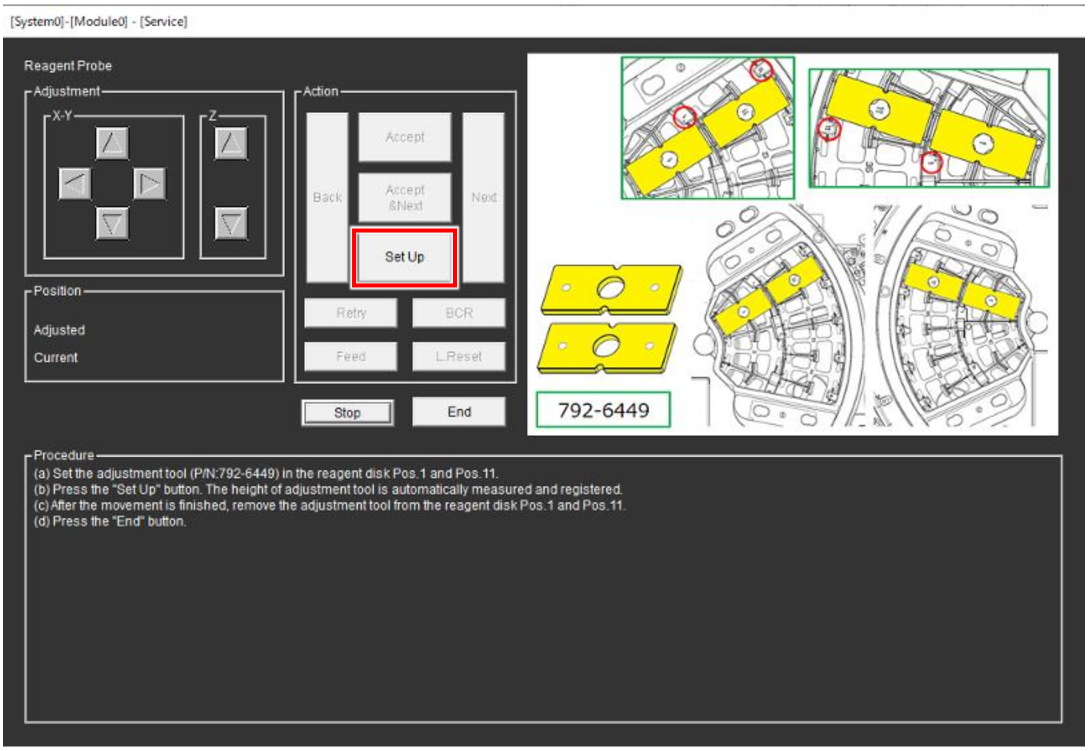


(2) Set the 4 tools (a) to Pos.1 and Pos.11 of the Reagent Disk A and B.



(a): BOTTLE PLATE C (P/N: 792-6453)

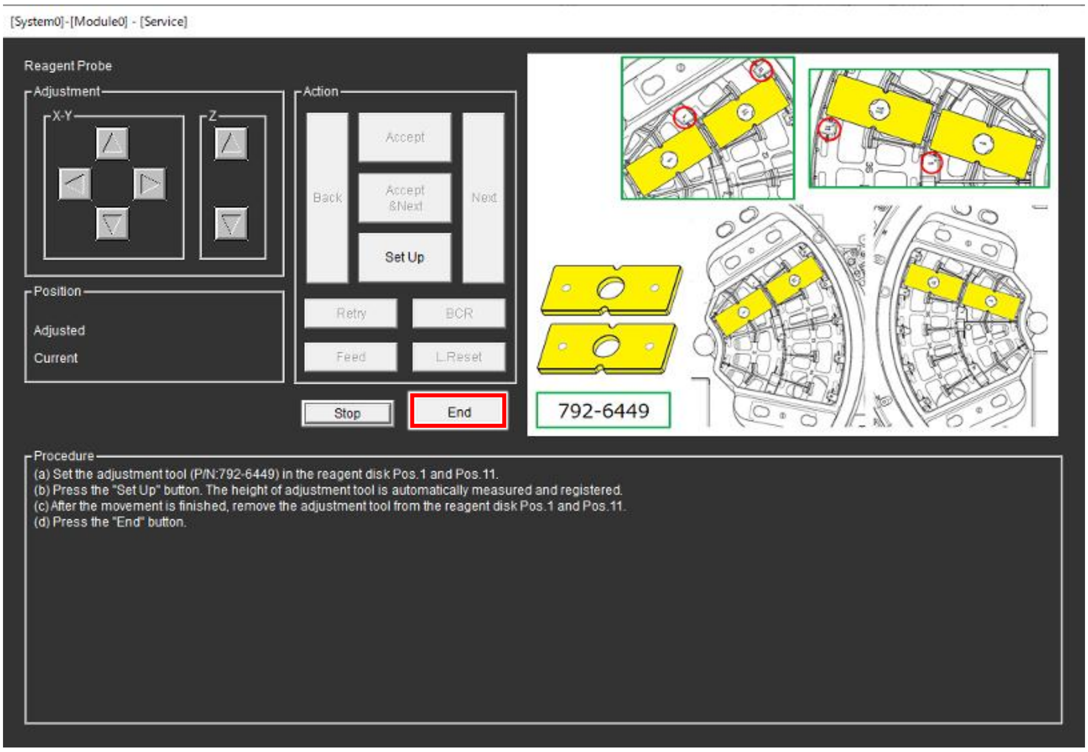
(3) Select [Set Up].



(4) Wait for Reagent Probe Vertical to end.

(5) Remove the 4 tools from Pos.1 and Pos.11 on the Reagent Disk A and B.

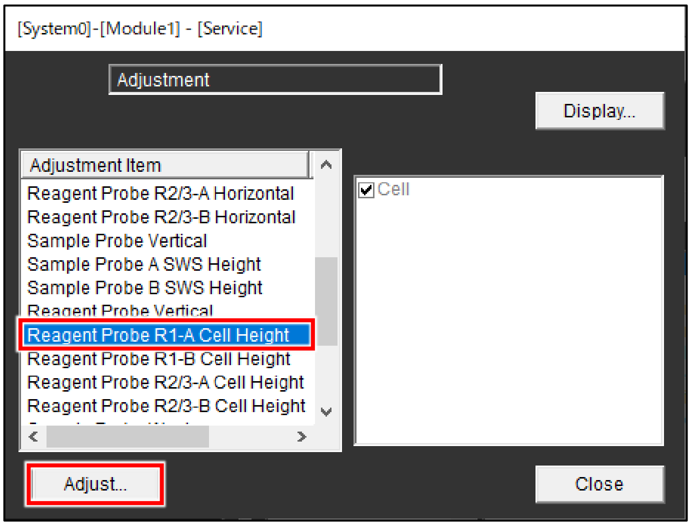
(6) Select [End].



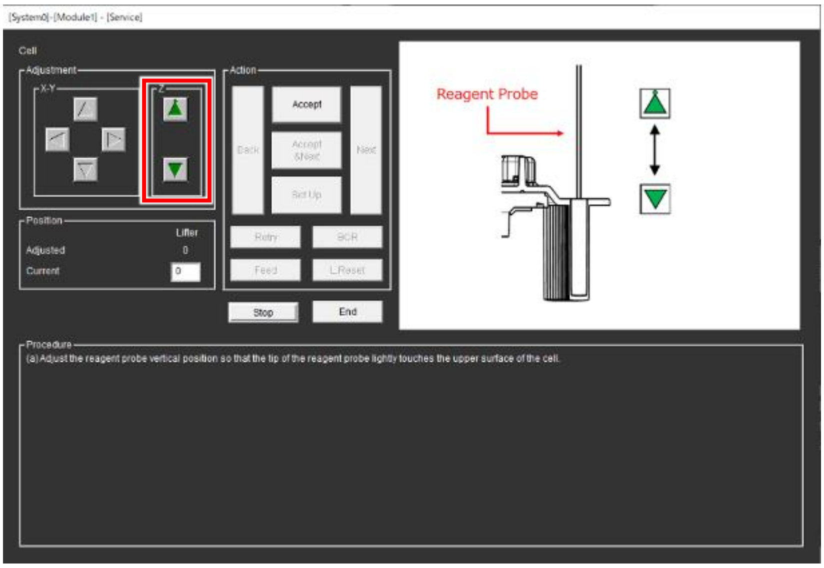
8. Reagent Probe R1-A Cell Height

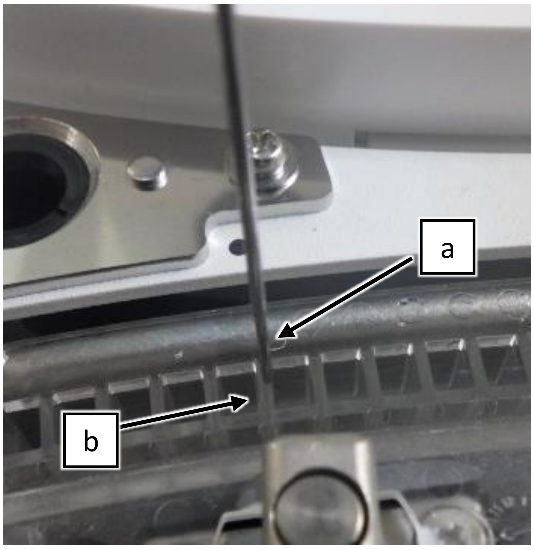
The adjustment procedure for R1-B, R2/3-A, R2/3-B are same as the procedure for R1-A.

(1) Select [Reagent Probe R1-A Cell Height] from Adjustment item and select [Adjust...].



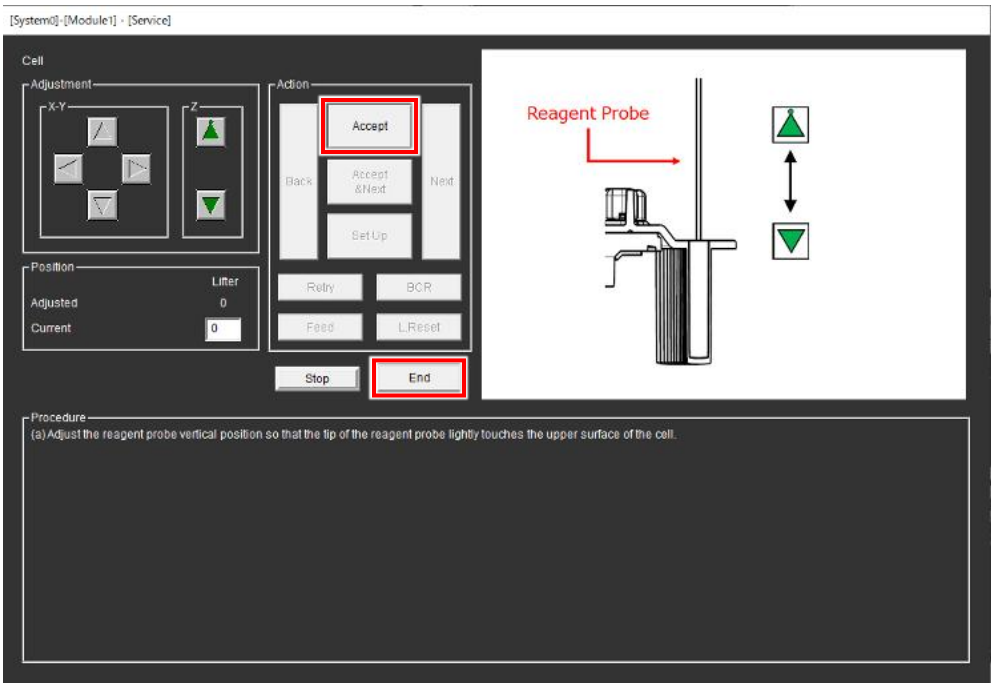
(2) Adjust Reagent probe height so that the tip of Reagent probe touches the upper surface of Cell (b) lightly using the 2 arrow buttons.





(3) Select [Accept].

(4) Select [End].



9. Reagent Probe R1-B Cell Height

The adjustment procedure for R1-B, R2/3-A, R2/3-B are basically same as the procedure for R1-A.

Perform the adjustment refer to the chapter 8.

10. Reagent Probe R2/3-A Cell Height

The adjustment procedure for R1-B, R2/3-A, R2/3-B are basically same as the procedure for R1-A.

Perform the adjustment refer to the chapter 8.

11. Reagent Probe R2/3-B Cell Height

The adjustment procedure for R1-B, R2/3-A, R2/3-B are basically same as the procedure for R1-A.

Perform the adjustment refer to the chapter 8.

12. Reagent Probe R1-A Recognize

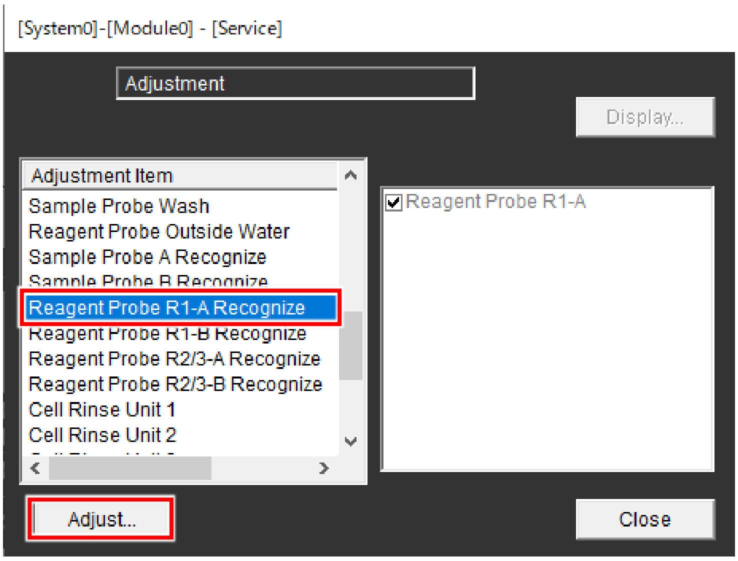
The adjustment of horizontal stop position for the Reagent Probe must be completed.

The adjustment of vertical position for the Reagent Probe must be completed.

If they haven't been adjusted yet, perform the adjustments in accordance with the adjustment procedure.

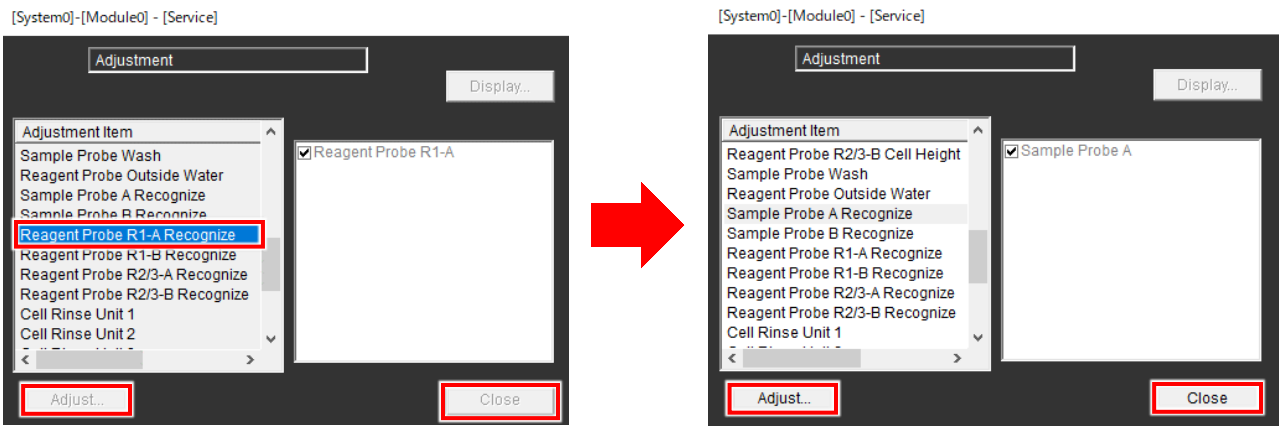
(1) Select [Reagent Probe R1-A Recognize] from the Adjustment item and select [Adjust...].

(2) The adjustment procedure for R1-B, R2/3-A, R2/3-B is basically same as the procedure for Reagent Probe R1-A.



(3) Wait for Reagent Probe R1-A Recognize to end.

It takes approximately 2 minutes.



(4) Make sure that any alarm does not occur in SSW and on CU.

If alarms occur, perform the section 3.7 and the chapter 7.

13. Reagent Probe R1-B Recognize

The adjustment procedure for R1-B, R2/3-A, R2/3-B are basically same as the procedure for R1-A.

Perform the adjustment refer to the chapter 12.

14. Reagent Probe R2/3-A Recognize

The adjustment procedure for R1-B, R2/3-A, R2/3-B are basically same as the procedure for R1-A.

Perform the adjustment refer to the chapter 12.

15. Reagent Probe R2/3-B Recognize

The adjustment procedure for R1-B, R2/3-A, R2/3-B are basically same as the procedure for R1-A.

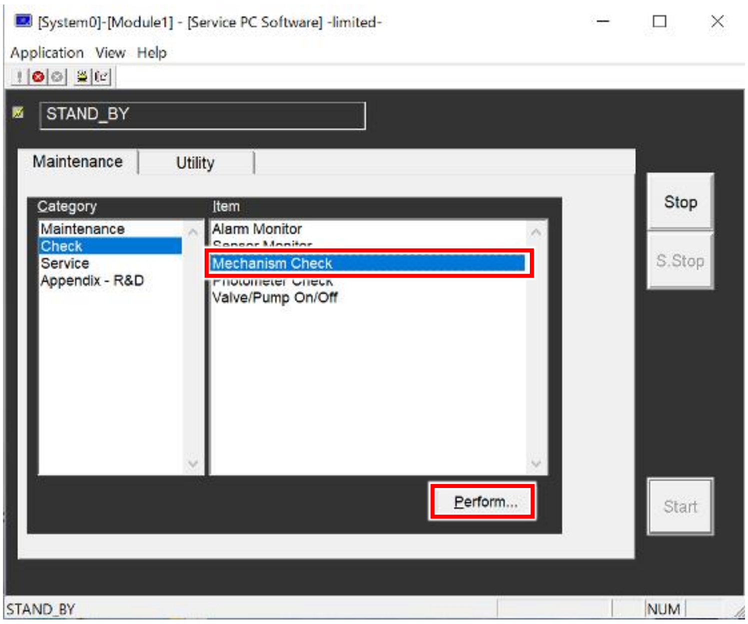
Perform the adjustment refer to the chapter 12.

16. Accomplishing the adjustment

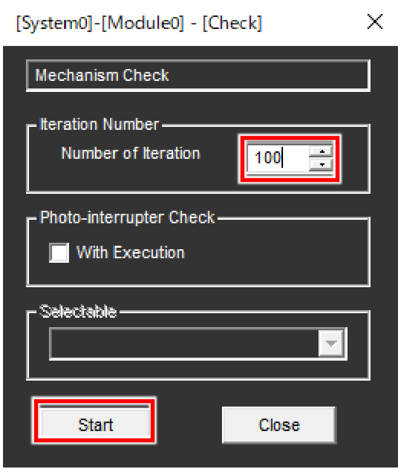
(1) Attach 2 Reagent Pipetting Covers.

(2) Attach Cell cover.

(3) Select [Check] > [Mechanism Check] > [Perform...].



(4) Set the number of iterations to 100 and select [Start].



(5) During Mechanism Check, make sure that Reagent probes do not contact anything.

If they contact something, retry the adjustments.

(6) Turn off Maintenance switch.

(7) Attach COVER MSW ASSY.

(8) Close TOP COVER F ASSY.

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